Time series analysis – 1

Agenda →

- Recap
  - generate forecasts
    - Simple models
      - naive
      - mean
      - seasonal Naive
      - Drift
    - → implementation
  - → Simple Expo Smoothening
    - Simple
    - Double
    - Triple

Time series →
analysis → missing values →
↓ ↳ outliers →
Trend, seasonality

FYI → Basic

Time series data

decomposition

additive model

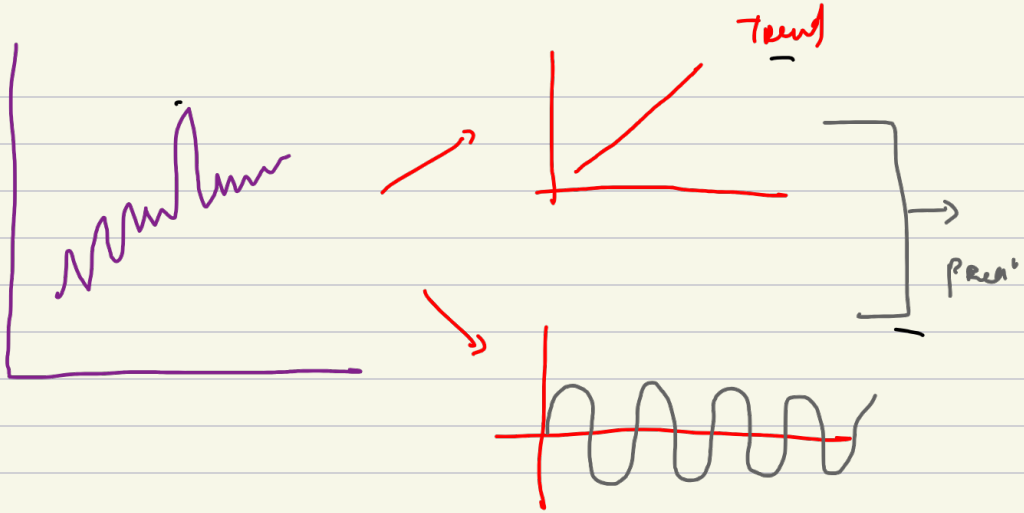
Trend + Seasonality → pred

Residuals → X

$\hat{y} - y$ = residual

↓

Hiccups



Residuals

$$y(t) = b(t) + s(t) + e(t)$$

$$e(t) = y(t) - [b(t) + s(t)]$$

Pred'

Agenda → MAPE < 5%

Algo → Classification → Accuracy, f1 Score, Precision, Recall

Regression → R^2 Score, MSE

Advanced & Supervised

DATE sales

Business =>

Unsupervised

R^2 → X X

MSE → 0 , 10000

MAPE → X

use it with Regression

$$\frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \Rightarrow \%$$

loss func^n X

MSE

Classification → F1 X

IRS X

Recall X

eg → actual sales = 350

pred^n = 315

MAPE = $\left|\frac{350 - 315}{350}\right| = \frac{1}{10}$

MAPE ⇒ 10%

Prediction = 385
Actual = 350
⇒ MAPE = 10%

Deviation

5% | 10% | 2%

Runderforecast , Poverforecast

MAPE → MPE Mean Percentage error

Agenda

MAPE < 5%

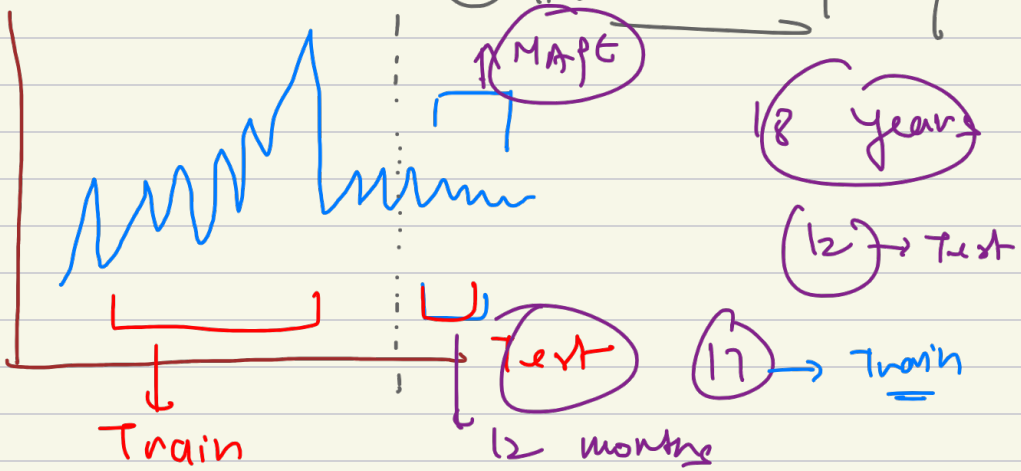
✗ Normal Regression ,

Time series

→ Random splitting ✗

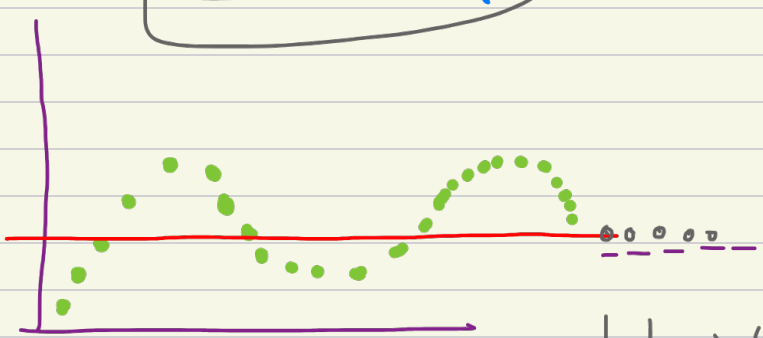
order

* Time Based splitting ⇒



1. Simple forecasting

1. Mean forecasting ⇒



t (t+1) (t+2) (t+3)

actual

Class

13 14 11 17

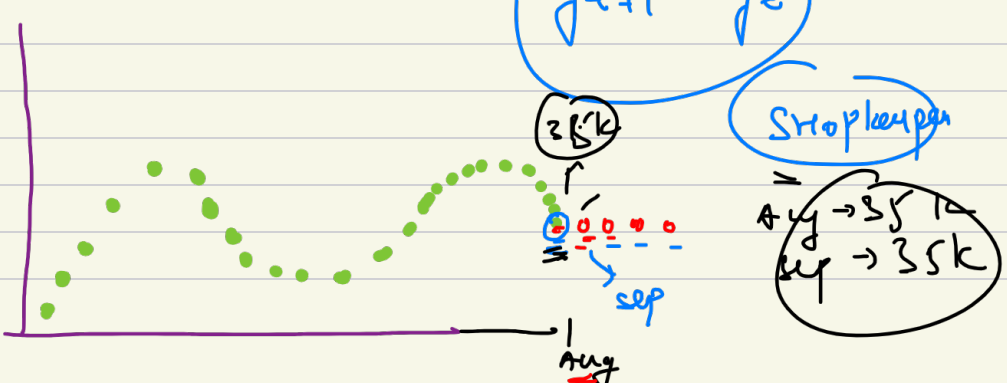
well educated guess

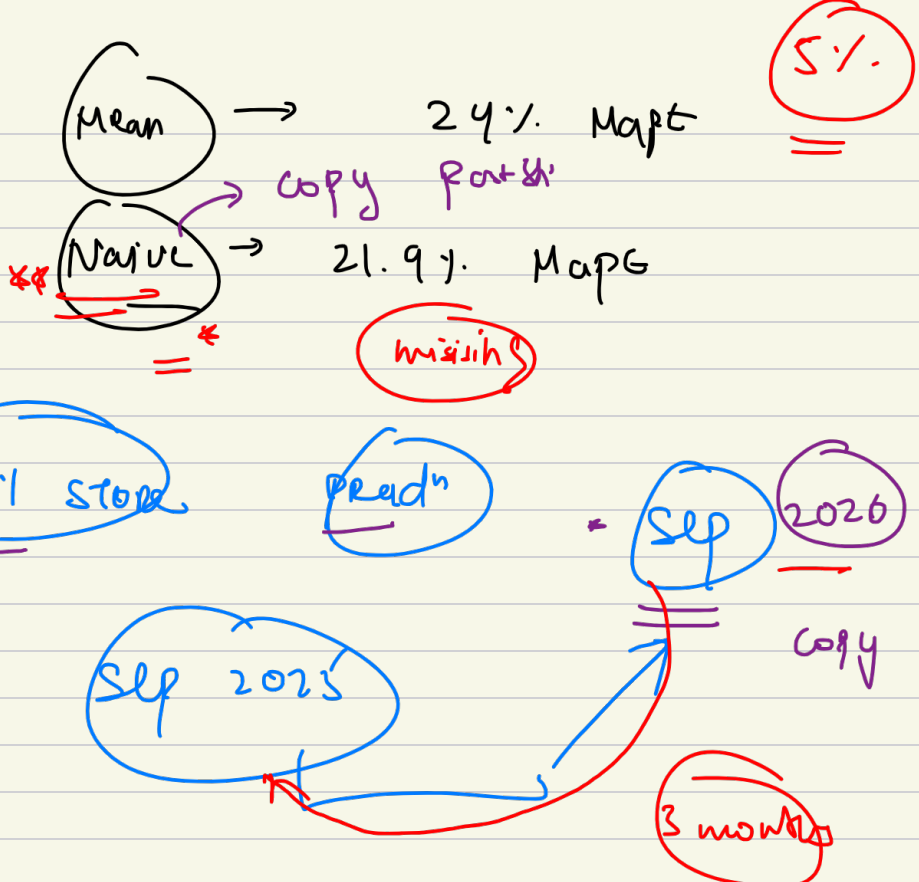
$$\hat{y}_{t+1} = \frac{y_1 + y_2 + y_3 + \cdots y_t}{t}$$

(t+2)
(t+3)
] pred^{n} remain same

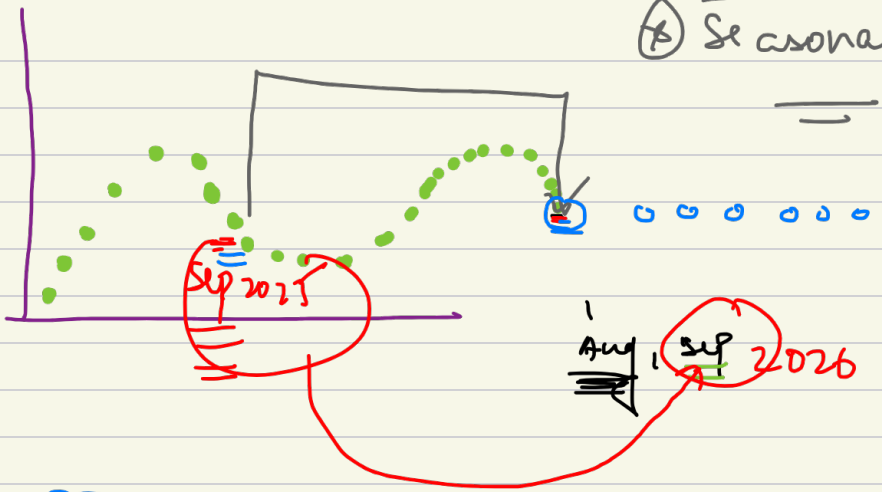
Naive approach

$$\hat{y}_{t+1} = y_t$$




Mean
24%. MAPE
5%.
Naive
copy Pattern
21.9%. MAPE
missing
Retail store
Predn
Sep
2026
copy
Sep 2025
3 months
Seasonality
Seasonal Naive

④ Seasonal Naive



$y_{t+h} = y_{t-T+h}$

T = 30
monthly
weekly
T = 7

current month
Aug

horizon
(1, 2, 3)
Sep
Oct
NOV

2026
Aug - 12 + 1
Aug 2025

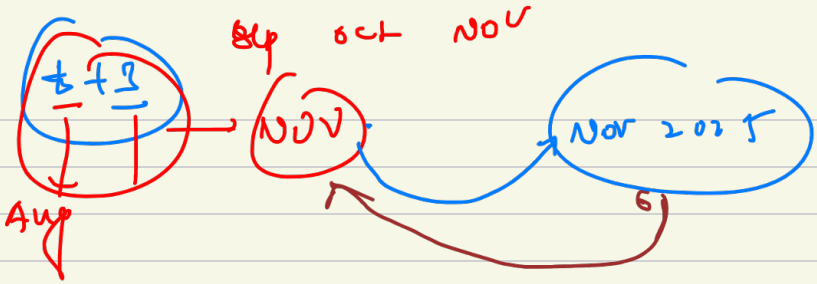
T = 12 ⇒

= Sep 2025 1 year = 12

8. 10:10

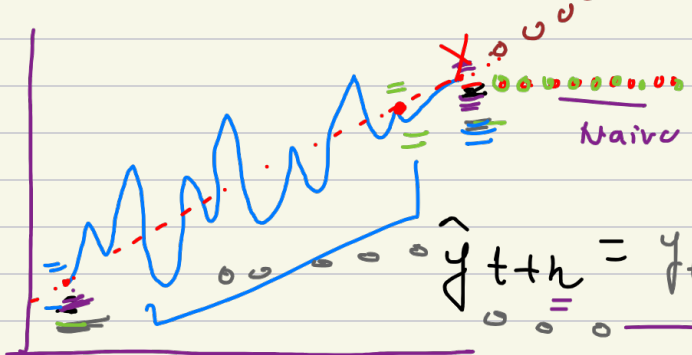
$\hat{y}_{Sep}$

Aug
(t+1) → Sep
(t+2) → Oct
(t+3) → NOV



DRIFT METHOD

slope



slope

$\hat{y}_{t+h} = y_t + h \cdot \text{slope}$

y_t → current month

$\frac{y_t - y_0}{t} = \frac{\Delta y}{\Delta t}$

$\Delta y \rightarrow$ Sep

$t = 1$

$t = \Delta y$

Sep → $t+1$

$\hat{y}_{t+1} = y_t + 1 \cdot \text{slope}$

Sep

$$\hat{y}_{t+2} = y_t + 2 \cdot \text{slope}$$ ← out

(1) get the slope

$$\frac{\text{Recent month} - 1\text{st value}}{\text{no. of pp}} = \text{slope}$$

$t = 1, 2, 3, \ldots 12$

problem → very sensitive to last pp

$$\text{slope} = y_t - y_0$$

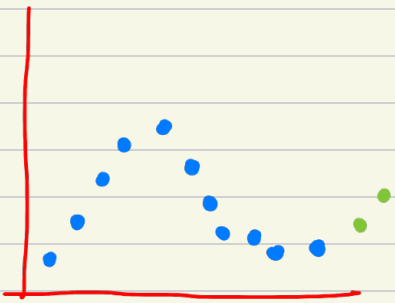
simplest methods

Moving Averages (MA)

window

window = 3

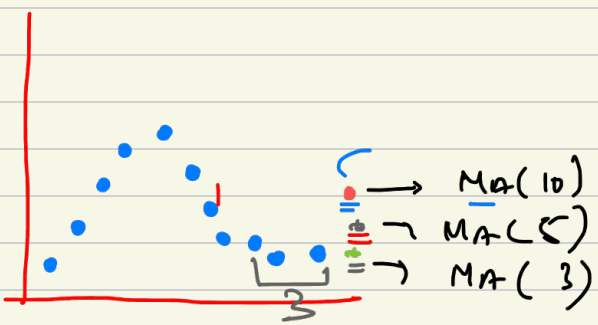
Moving avg



window → 10 → 5

3

window → 9



MA → window

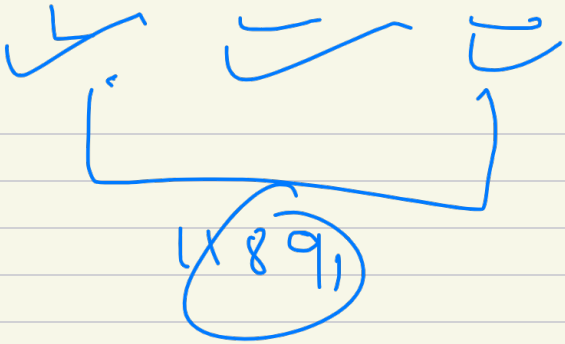
```
11524.        , 11053.        , 11766.        , 12921.        ,
 8927.        , 10083.        , 11151.        , 10863.        ,
11788.        , 12713.        , 11283.        , 12605.        ,
10970.        , 12194.        , 12393.        , 13360.        ,
 9833.        , 10690.5       , 11548.        , 11935.        ,
13655.        , 12825.        , 15164.6       , 12899.        ,
11936.        , 12981.        , 13563.5       , 14146.        ,
 9997.        , 10501.        , 11496.        , 12416.        ,
13232.        , 13343.        , 12402.        , 12336.        ,
12270.        , 13257.        , 11907.        , 14752.        ,
 9839.        , 10499.        , 12161.        , 12555.5       ,
12950.        , 14343.        , 13308.        , 12666.        ,
12758.33333333, 12850.66666667, 12943.        , 15164.6       ,
10089.        , 11762.        , 13549.        , 12859.        ,
13677.        , 15118.        , 14630.        , 14142.        ,
13713.        , 13342.        , 14455.        , 15164.6       ,
10485.        , 11891.        , 13921.        , 14453.5       ,
14986.        , 15164.6       , 12856.        , [illegible]   ,
13349.        , 14150.        , 14698.        , 15164.6       ,
11033.        ])
```

window = 3

→ train

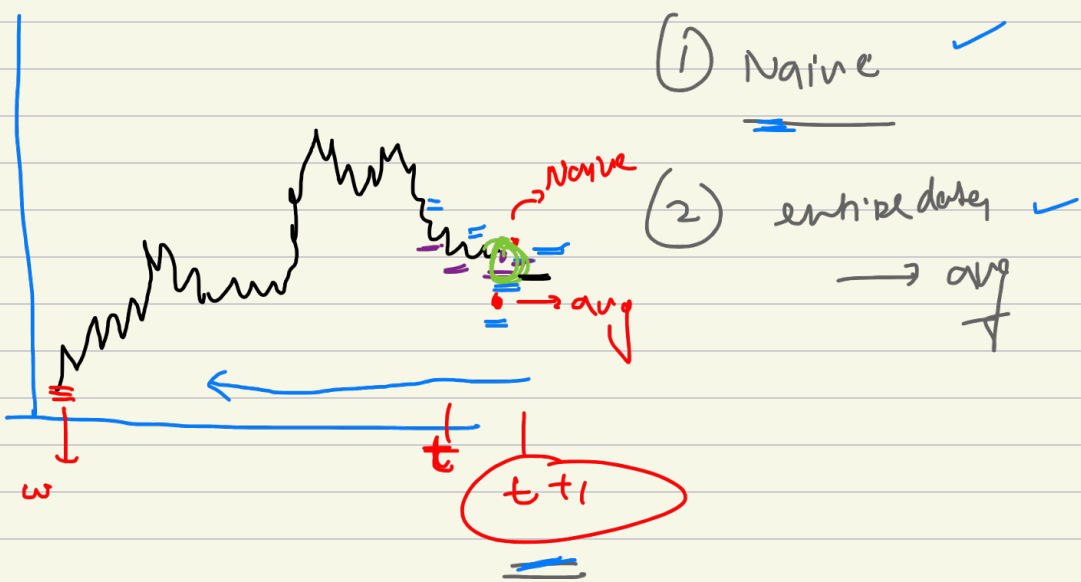
moving avg





Simple Exponential Smoothing

(*) [Naive + mean + window]



Trust more → Recent value

2nd most Recent → slightly less

3rd most →

least amount of Trust → 1st value

(weighted)

↓

wts ↓ as u go past in the time

Highest wt

$$\hat{y}_{t+1} = w_1 y_t + w_2 y_{t-1} + w_3 y_{t-2} + \dots + w y_0$$

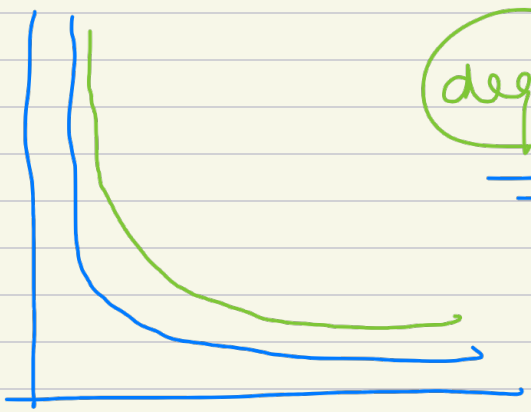
least wt.

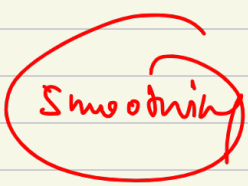
How to get these wts?

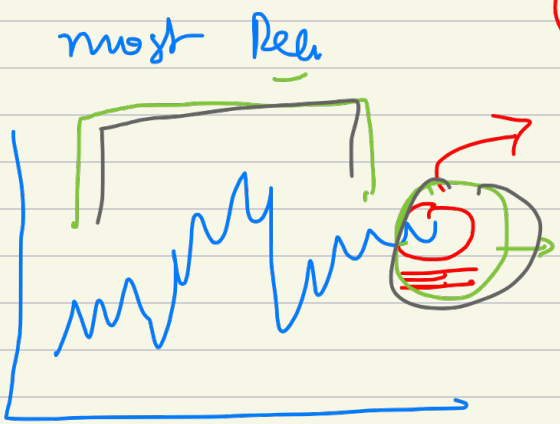
func^n → exponential smoothing

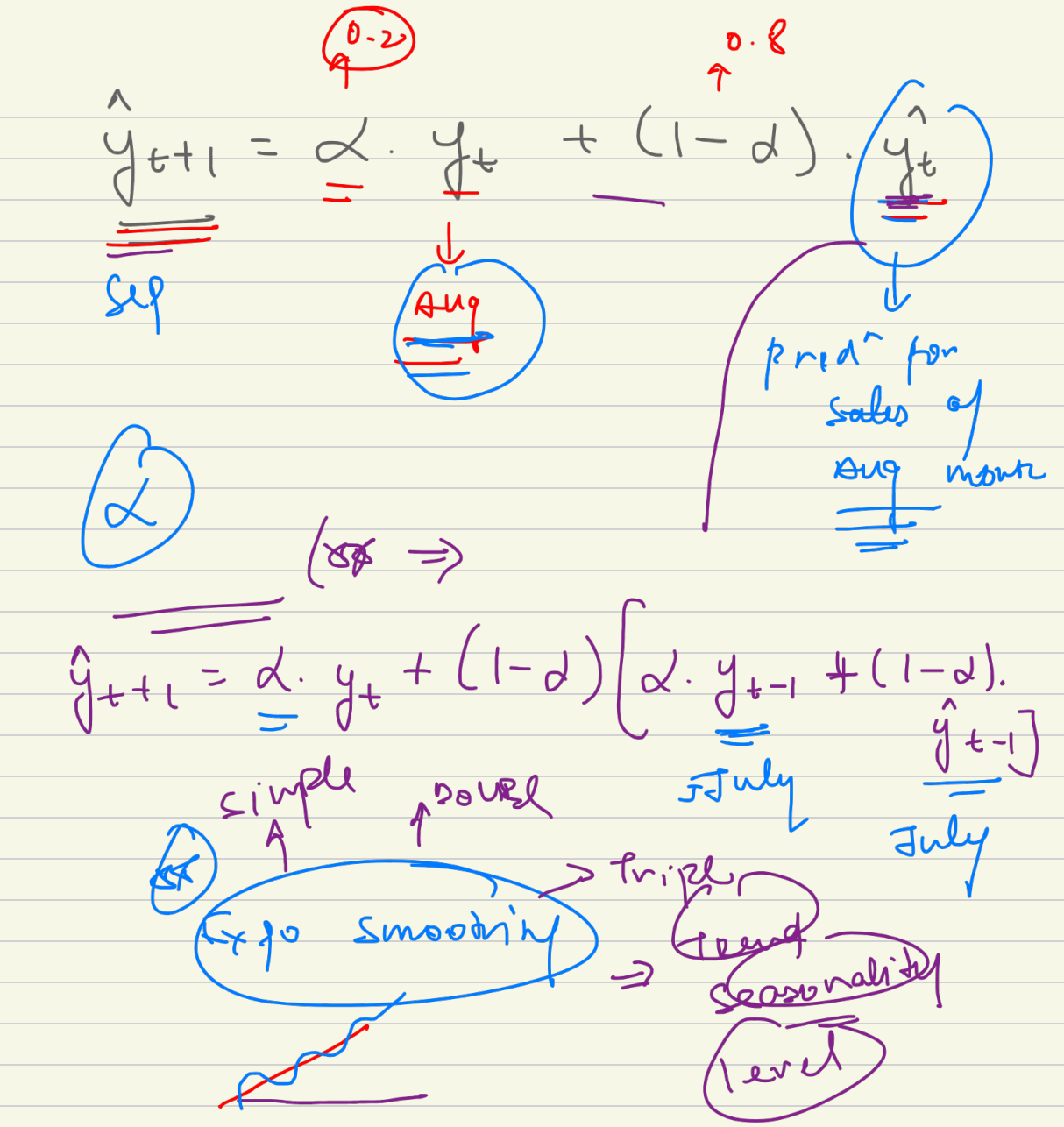
weights ↓ exponentially


most percent
wt ≈ 0.0.001


degree


Smoothing


most Rel


0.2
0.8
$\hat{y}_{t+1} = \alpha \cdot y_t + (1-\alpha) \cdot \hat{y}_t$
Sep
Aug
pred^ for Sales of Aug month
α
$\hat{y}_{t+1} = \alpha \cdot y_t + (1-\alpha)\left[\alpha \cdot y_{t-1} + (1-\alpha) \cdot \hat{y}_{t-1}\right]$
July
July
simple
double
triple
Exp. Smoothing
trend
seasonality
level

Quiz time!

Time Left: 37s

Forecasts are said "Naive" when:

25 users have participated

A	They are exact same as the last value from the training data	88%
B	They are short-term	0%
C	They are long-term	4%
D	They predict incorrect forecasts	8%

End Quiz Now

Quiz time!

Time Left: 24s

Statement I: Naive method is capable of capturing seasonality. Statement II: Forecasting with mean/median value will give a flat forecast around mean/median value Statement III: Seasonal Naive is capable of capturing seasonality as well as trend. Which of the above statements is/are true?

13 users have participated

A	Statement I	8%
B	Statement II	69%
C	Statement III	8%
D	All of them are true	15%

End Quiz Now

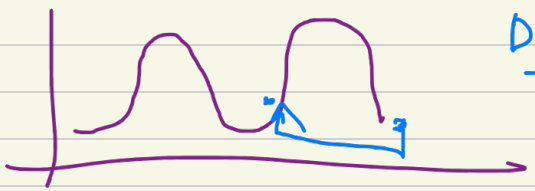
Doubts

typo

Smoothing - level = α = 20% (0.2)

$(1-\alpha)$ = 80%

Seasonal Noise → seasonality ⇒



Drift means → trend

DT ⇒

Seasonal + Drift

DATA

→ Mathematics

DATA

TREE

implementing